

ME301 Lab 1 Report

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ABSTRACT

This lab investigated reactive and feedback control strategies using a hexapod robot. Reactive control was implemented to enable obstacle avoidance based on ultrasonic sensing and feedback control was used to achieve wall-following capabilities at a fixed distance. Experimental results demonstrate trade-offs between complexity, speed, and robustness across control strategies, and highlight the importance of sensor feedback and controller tuning to achieve stable, autonomous robot behavior.

I. INTRODUCTION

Control is a fundamental component of robotics that governs how a robot senses its environment, interprets sensor data, and executes responsive actions. Without effective control, a robot may move unpredictably, drift off course, collide with obstacles, or fail to complete even simple tasks reliably. In real-world robotic systems, such as autonomous vehicles, warehouse robots, and legged robots operating on uneven terrain, robust control is essential for maintaining stability, accuracy, and safety. As robotic systems grow in complexity and are deployed in increasingly unstructured environments, the role of effective control strategies becomes progressively more important.

This lab explores two fundamental forms of control: open-loop control and closed-loop (feedback) control. Closed-loop control uses sensor measurements to achieve and maintain a desired system state by continuously correcting for error [1, p. 121]. Examples of feedback control outside of robotics include thermostats and autonomous vehicle navigation systems, where sensor data is used to guide decisions toward a target condition. In robotic applications, feedback control may be used to achieve a goal, such as reaching a target position, or to maintain a goal, such as balancing upright while walking.

Feedback control commonly incorporates proportional (P), integral (I), and derivative (D) components. These controllers operate based on system error, defined as the difference between the desired and current states. Proportional control generates an output proportional to the instantaneous error, integral control accounts for the accumulated error over time, and derivative control responds to the rate of change of error [1, p. 128]. In practice, these components are often combined to balance responsiveness and stability in applications such as robotic arms, mobile robots, and self-balancing platforms.. A key challenge in implementing feedback control is controller tuning, as poorly chosen gains can lead to oscillations or instability.

In contrast, open-loop control, also referred to as feedforward control, operates without sensor feedback. This approach relies on predefined actuation commands or trajectories to drive robot behavior. While open-loop control is computationally simple and efficient, it is highly sensitive to disturbances, environmental variation, and modeling inaccuracies due to the absence of corrective feedback.

Reactive control is another control strategy explored in this lab. Reactive controllers directly map sensor inputs to actions, enabling robots to respond quickly to environmental stimuli. For example, upon detecting an obstacle, a robot may immediately execute a turning maneuver to avoid collision. This approach is widely used in mobile robots and autonomous platforms where fast response is more critical

than long-term planning. While reactive control is effective for real-time responsiveness, it presents scalability challenges. As the number of environmental conditions increases, multiple behaviors may be triggered simultaneously, requiring techniques such as action fusion or arbitration to resolve conflicting actions.

Selecting an appropriate control strategy is critical to a robot's functionality and performance. The robot's objectives, operating environment, and actuation and computational constraints must all be considered during controller design. Overly complex control solutions can introduce unnecessary instability or inefficiency, particularly when simpler approaches are sufficient.

This lab investigates feedback and reactive control through the implementation of walking and turning behaviors, obstacle detection, reactive obstacle avoidance, and wall-following using feedback control. These tasks highlight the trade-offs between simplicity and robustness in robot control and demonstrate how sensing, control, and actuation must be tightly integrated to enable effective autonomous navigation.

II. METHODS

The robot used for this lab was the hexapod from the previous lab. To completely optimize the walking, the tripod gait from the previous assignment was selected and modified. The tripod gait was chosen as it provided the greatest forward speed compared to the wave and ripple gaits.

To optimize for speed, the `time.sleep()` values for each servo step was minimized from 0.5 seconds to 0.2 seconds. This modification was motivated by the expectation that decreasing step delay would increase the walking frequency, resulting in greater distance traveled over time while maintaining acceptable stability. The experiment aimed to determine whether the increased speed introduced additional drift or degraded straight-line motion.

The optimized tripod gait was tested five times for each of 5, 10, and 15 second durations. For each trial, the final position and total distance traveled were recorded. Final x and y positions were measured using a meterstick, where the positive x-direction was defined as the robot's forward direction and the positive y-direction as the robot's right side. These measurements were collected to evaluate both the speed and directional consistency of the gait and to determine whether drift accumulated differently over short versus long time durations.

After optimizing forward walking, three turning gaits were developed: a left turn, a right turn, and a 180-degree turn. These turning gaits were designed to act as simple motion building blocks that could later be reused for reactive control. A left turn was implemented by raising and rotating Group 1 of the tripod gait fully counterclockwise, grounding Group 1, and then raising and rotating Group 2 counterclockwise while Group 1 returned to its home position. This sequence was repeated four times to achieve a 90-degree rotation. The right turn was implemented identically, with the rotation directions reversed. The 180-degree turn consisted of executing the right-turn gait twice consecutively.

Each turning gait was tested five times, during which both the change in heading and execution time were measured. Heading change was measured using a digital compass placed on the robot chassis, while execution time was measured with a stopwatch. These measurements were chosen to evaluate the accuracy and efficiency of each turning behavior. Specifically, the heading change quantified whether each turn reliably achieved the intended angle, while execution time revealed trade-offs between speed and rotational precision.

These gaits are representative actions in reactive control. A left turn was triggered when obstacles were detected in front of and to the right of the robot, a right turn when obstacles were detected in front of and to the left, and a 180-degree turn when obstacles were detected on all three sides. Obstacles were classified as detected when the sonar sensor measured a distance less than 350 mm from the robot chassis. Obstacle detection was performed using an ultrasonic sonar sensor mounted on top of the robot chassis, as shown in Figure 1.

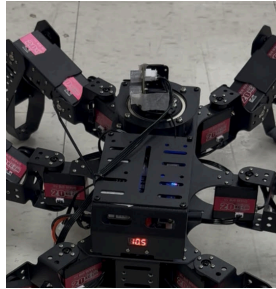


Figure 1. Sonar Sensor on Top of Robot Chassis

The sonar sensor was mounted on a rotating servo, allowing the robot to actively scan its environment. To reduce unnecessary scanning and improve responsiveness, the reactive algorithm first checked for obstacles directly ahead. If no front obstacle was detected, the robot continued forward motion. If a front obstacle was detected, the sonar rotated left and then right to determine obstacle presence on each side. This structured scanning approach was chosen to minimize sensor actuation while still gathering sufficient information to make reliable navigation decisions.

This reactive control algorithm was tested by placing a hand in front of the sensor to activate a certain action. For example, a hand could be placed when the sensor checks forward, left, and right to activate a turn180() action.

Finally, a feedback control algorithm was implemented to enable wall-following behavior, which represents a maintenance control goal. The robot was required to walk forward while maintaining a fixed distance of 35 cm from a wall. Depending on the wall location, the sonar sensor was first rotated to face the wall. The controller computed an error as the difference between the measured sonar distance and a desired distance of 380 mm, which accounted for sensor placement offset. A deadband of ± 15 mm was included to reduce high-frequency oscillations near the desired distance.

A proportional (P) controller was used, producing a control output proportional to the distance error with gain K_p . The output of the controller was implemented as a shift applied to the default swing amplitude

of the tripod gait. By increasing or decreasing the leg swing on one side of the robot relative to the other, the robot adjusted its trajectory toward or away from the wall. This method was chosen to preserve continuous forward motion while enabling smooth lateral correction.

The gain K_p was initially set to 1 and iteratively adjusted based on observed performance. Overcorrection indicated an excessively high gain, while insufficient correction suggested a low gain. Through experimental tuning, a value of $K_p = 0.1$ was selected to balance responsiveness and stability, allowing gradual correction without inducing oscillations.

The feedback-controlled gait was tested five times for each of 5, 10, and 15 second durations. The total distance traveled and the final distance from the wall were measured using a meterstick. For each trial, the robot was placed at a starting distance of 35.5 cm from the wall. The wall used in all trials was a row of lockers. Total distance was measured to evaluate whether feedback correction reduced overall forward efficiency, while wall distance was measured to assess the controller's ability to maintain the desired setpoint. Multiple durations were tested to determine whether the system converged toward a steady-state wall distance over time and to observe how transient corrections influenced short versus long trials. An iteration of the wall following can be seen in Figure 2 below.

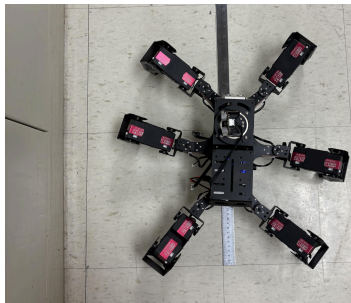


Figure 2. Example of Wall Following Trial

III. RESULTS

The tripod gait distances for 5, 10, and 15 seconds are recorded below in Table 1. As mentioned previously, the positive x direction is the forward direction of the robot while the positive y direction is the right side of the robot. Each time duration was repeated 5 times.

Table 1. Tripod Gait Distances for 5, 10, and 15 Seconds

Time (sec)	X Displacement (cm)	Y Displacement (cm)	Total Distance (cm)
5	59.0, 57.5, 63.0, 58.5, 64.5	-1.1, -0.9, -2.5, -0.9, -2.5	59.1, 57.5, 63.0, 58.5, 64.5
10	144, 138.5, 150.5, 151, 147	-1.1, -2.4, -1.9, -1.7, -1.4	144.0, 138.5, 150.5, 151.0, 147.0
15	173.5, 172, 168, 180, 175.5	-9.0, -11.0, -11.5, -10.6, -12.0	173.7, 172.3, 168.4, 180.3, 175.9

The change in heading and the time of execution of each of the 3 turning gaits are recorded below in Table 2. Each turning gait was repeated 5 times.

Table 2. Turning Gait - Change in Headings and Time of Execution

Type of Turn	Change in Heading (deg)	Time of Execution (sec)
Left	90.1, 90.0, 90.0, 89.8, 90.0	2.46, 2.81, 2.69, 2.51, 2.83
Right	90.0, 89.8, 89.9, 90.0, 90.1	2.74, 2.68, 2.54, 2.43, 2.66
180 Degree Turn	180.2, 180.3, 179.7, 180.0, 180.0	4.79, 5.11, 4.83, 4.81, 4.87

The walking distance and the distance from the wall for the tripod gait with sensor feedback is recorded below in Table 3.

Table 3. Tripod P-Control Feedback Gait - Walking Distances and Final Distance from Wall

Time (sec)	X Displacement (cm)	Distance from Wall (cm)
5	61.0, 62.5, 62.0, 61.5, 61.0	35.2, 34.2, 34.0, 35.6, 35.2
10	111.4, 115.5, 113.5, 116, 115.5	39.6, 38.4, 38.9, 39.5, 39.0
15	181.0, 176.5, 182.5, 178.0, 180.5	38.0, 40.9, 38.3, 39.6, 38.7

IV. DISCUSSION

The optimized tripod gait demonstrated that increasing stride frequency by decreasing servo movement times significantly improved forward speed, but also introduced a consistent leftward drift over time. As shown in Table 1, while the robot generally maintained forward motion along the positive x-direction, deviations in the negative y-direction became more pronounced as trial duration increased. This behavior suggests that small asymmetries in servo response, ground contact, or mechanical alignment accumulated over time during open-loop walking. In particular, minor discrepancies between the left and right leg motions could cause one side of the robot to generate greater effective propulsion. Additionally, friction pads on the legs may have accumulated dust throughout testing, leading to inconsistent traction across trials. Although small variations in initial robot placement may have contributed to trial-to-trial variability, the consistent leftward drift observed across all trials indicates a systematic imbalance, most likely originating from servo output or leg geometry.

The turning gaits revealed a clear trade-off between rotational accuracy and execution time. While the left, right, and 180-degree turns produced heading changes close to their intended angles, minor overshoot and undershoot were observed across trials. These deviations likely resulted from uneven leg-ground friction and slippage during rotation, particularly as the robot relied on repeated tripod cycles rather than continuous rotation. Because these turning behaviors were designed for reactive control

rather than precise orientation control, this level of accuracy was sufficient for obstacle avoidance tasks, where approximate directional changes are acceptable.

The reactive obstacle avoidance successfully demonstrated rapid sensor-to-action mapping without relying on control feedback or interpreting the system state. The robot consistently executed appropriate turning behaviors in response to different obstacle configurations, validating the reactive control logic. However, testing revealed that performance was sensitive to sensor noise and measurement timing. Because decisions were made based on instantaneous sonar readings, false positives occasionally triggered unintended actions. This highlights a fundamental limitation of reactive control: while highly responsive, it lacks robustness and contextual awareness. Future improvements could include averaging multiple sensor readings or incorporating short-term memory of past measurements to reduce the influence of unreasonable data.

The wall-following feedback controller showed that proportional control alone can be sufficient for maintaining a desired distance from a wall when properly tuned. With an appropriately selected gain and deadband, the robot converged toward the target distance without excessive oscillations. Over longer trials, the robot tended to stabilize near the desired setpoint, indicating steady-state behavior. However, purely proportional control exhibited limited corrective strength when large disturbances occurred, resulting in slow or incomplete recovery. This suggests that incorporating integral or derivative control terms could improve long-term accuracy and transient response, though at the cost of increased system complexity and tuning effort.

A significant source of error in the feedback controller arose from the incident angle of the sonar sensor relative to the wall. As the robot made corrective turns, the sensor was no longer parallel to the wall surface, leading to distance readings that underestimated proximity. This occasionally caused the robot to remain closer to the wall than indicated by the sensor or to overcorrect for visually small errors. In future implementations, repositioning the sonar sensor or adding multiple sensors could help ensure more reliable distance measurements and improve overall wall-following performance.

V. CONCLUSION

This lab explored the implementation and evaluation of reactive and feedback control strategies on a hexapod robot. The non-feedback walking and turning gaits provided fast and reliable motion, but exhibited minor drift and inconsistencies due to environmental conditions. The reactive control enabled effective obstacle avoidance through sensor to action mapping, demonstrating strong responsiveness but limited robustness to noise. Feedback control using a proportional controller allowed the robot to maintain a desired wall distance and converge towards a stable behavior over time.

Overall, these results illustrate the importance of sensor feedback and controller tuning in autonomous robotics, as well as the trade-offs between complexity, speed, and robustness. This lab demonstrates the importance of analyzing a robot's goals, limitations, and choosing a control strategy that achieves reliable robot navigation in real-world environments.

REFERENCES

- [1] “The robotics primer,” Choice Reviews Online, vol. 45, no. 06, pp. 45–322245–3222, Feb. 2008, doi: <https://doi.org/10.5860/choice.45-3222>.